

INNER WHALEBACKS, NT, Canada

WMO: 711620

Lat: 61.921N

Lon: 113.726W

Elev: 165

StdP: 99.36

Time Zone: -7.00 (NAM)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.6	(c) -33.3	(d) -39.7	(e) 0.1	(f) -35.5	(g) -37.2	(h) 0.1	(i) -33.3	(j) 15.9	(k) -19.4	(l) 14.4	(m) -20.7	(n) 5.4	(o) 310	(p) 0.980

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.6	(c) 20.3	(d) 17.1	(e) 18.8	(f) 16.0	(g) 17.6	(h) 15.1	(i) 17.5	(j) 19.9	(k) 16.3	(l) 18.5	(m) 15.3	(n) 17.4	(o) 4.2	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.9	19.2	15.2	11.0	18.1	14.2	10.3	17.0	50.1	19.8	46.6	18.7	43.6	17.5	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	13.4	12.2	-35.0	23.8	2.6	2.0	-36.9	25.2	-38.4	26.4	-39.9	27.5	-41.8	29.0
			-35.4	19.2	2.6	2.5	-37.3	21.0	-38.8	22.4	-40.2	23.8	-42.1	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.4	-22.5	-21.1	-16.6	-6.8	1.6	7.9	13.6	14.0	9.1	2.0	-6.9	-16.4		
	DBStd	13.80	6.95	6.53	7.19	6.08	3.36	2.74	3.42	2.68	3.34	3.86	5.29	6.77		
	HDD10.0	5182	1008	870	825	503	261	76	7	2	55	248	508	819		
	HDD18.3	7946	1267	1103	1083	753	519	314	152	136	277	507	758	1078		
	CDD10.0	283	0	0	0	0	0	11	117	126	28	0	0	0		
	CDD18.3	6	0	0	0	0	0	0	4	2	0	0	0	0		
	CDH23.3	4	0	0	0	0	0	0	4	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.5	6.3	6.0	6.3	6.4	5.8	6.0	5.4	6.0	7.1	7.7	7.8	6.8		
(15) Precipitation	PrecAvg	344	17	12	14	12	22	38	51	55	44	33	28	18		
	PrecMax	447	30	27	29	19	43	94	98	95	81	61	45	30		
	PrecMin	231	4	3	4	4	3	5	18	22	12	12	16	7		
	PrecStd	41	7	5	7	4	10	20	18	19	18	11	8	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.4	-4.2	1.0	5.1	11.4	18.3	23.6	21.5	19.1	10.6	5.3	-0.8		
		MCWB	-2.7	-5.1	-0.2	3.1	7.0	12.5	19.6	17.5	18.7	8.6	3.7	-1.5		
	2%	DB	-7.1	-6.8	-1.0	3.6	8.2	15.1	21.3	19.6	15.0	9.3	2.9	-2.6		
		MCWB	-7.5	-7.2	-1.9	2.0	5.1	11.0	17.8	16.4	12.9	7.6	1.8	-3.3		
	5%	DB	-10.6	-9.7	-3.0	2.6	6.5	13.0	19.7	18.4	13.9	8.1	1.6	-4.6		
		MCWB	-10.9	-10.1	-3.7	1.3	4.0	9.9	16.6	15.8	12.1	6.7	0.7	-5.1		
	10%	DB	-13.2	-11.8	-5.7	1.7	5.3	11.4	18.3	17.4	13.1	6.8	0.0	-6.8		
		MCWB	-13.6	-12.2	-6.4	0.6	3.3	8.7	15.7	15.0	11.4	5.4	-0.6	-7.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.7	-5.0	-0.2	3.2	7.3	13.3	20.0	18.0	18.8	9.0	3.8	-1.5		
		MCDB	-2.4	-4.3	0.7	4.8	10.6	17.0	22.9	20.6	19.2	10.1	5.3	-0.9		
	2%	WB	-7.5	-7.2	-2.0	2.1	5.4	11.5	18.2	16.9	13.2	7.8	1.9	-3.2		
		MCDB	-7.1	-6.7	-1.1	3.3	7.7	14.5	20.8	19.0	14.5	9.1	2.8	-2.6		
	5%	WB	-11.0	-10.0	-3.7	1.4	4.2	10.2	16.9	16.1	12.4	6.7	0.8	-5.1		
		MCDB	-10.7	-9.7	-3.0	2.5	6.2	12.6	19.2	18.1	13.7	8.1	1.6	-4.6		
	10%	WB	-13.5	-12.2	-6.4	0.7	3.4	9.0	15.9	15.2	11.6	5.4	-0.7	-7.2		
		MCDB	-13.3	-11.9	-5.7	1.7	5.1	11.2	18.0	17.2	12.9	6.7	0.0	-6.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	8.9	9.3	6.4	6.3	5.6	3.6	3.2	3.2	4.1	5.2		
		MCDBR	6.9	8.0	9.7	8.2	7.5	9.0	6.9	4.8	3.9	3.7	4.7	6.6		
		MCWBR	6.7	7.9	9.1	7.0	5.2	5.9	4.4	3.2	2.9	3.3	4.4	6.4		
	5% WB	MCDBR	6.8	8.1	9.6	7.6	7.2	8.5	6.2	4.6	3.7	3.6	4.6	6.4		
		MCWBR	6.7	8.0	9.0	6.6	5.2	6.0	4.4	3.3	2.9	3.3	4.4	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.193	0.243	0.267	0.310	0.321	0.348	0.372	0.342	0.314	0.274	0.198	0.154		
	taud		2.066	2.225	2.262	2.321	2.491	2.424	2.412	2.500	2.544	2.425	2.067	1.937		
	Ebn at Noon		540	747	857	882	903	878	843	844	808	708	511	400		
	Edh at Noon		34	61	86	101	94	102	101	85	68	51	33	20		
(44) All-Sky Solar Radiation	RadAvg		0.30	1.11	2.87	4.88	5.89	6.25	5.56	4.18	2.52	1.01	0.30	0.14		
	RadStd		0.03	0.09	0.18	0.26	0.37	0.50	0.27	0.20	0.19	0.10	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	+1.41	N/A	N/A	+0.48	N/A	N/A	N/A	N/A

Nomenclature: See separate page